The best constant in Smale's mean value problem exceeds 1.048

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Abstract

For a complex polynomial p of degree $d \geq 2$ and a point z with $p'(z) \neq 0$ let $R(p,z) = \min_{p'(c)=0} |p(z)-p(c)|/(|z-c|\,|p'(z)|)$, and let $K_d = \sup R(p,z)$ over polynomials of degree d. Smale (1981) proved $K_d \leq 4$ and conjectured $K_d \leq 1$ (strong form $K_d = (d-1)/d$). A non-explicit disproof of $K_d \leq 1$ for some (very large) d was announced in September 2026. We prove that K_d is non-decreasing and that $\lim_{d\to\infty} K_d \geq 1.048$; thus no constant below 1.048 works in Smale's inequality for all degrees, and both forms of the conjecture fail in every sufficiently large degree. The proof has two parts. (i) An explicit *continuum witness*: a rounded Koebe map Φ with Gaussian-rational parameters and a Blaschke-type function F on the domain $\Phi(\mathbb{D})$, whose defining properties — injectivity of Φ, absence of critical points of F, and the boundary inequality $|F(z)| \geq 1.048\,|z-z_0|$ — are *formally verified in Lean 4/Mathlib*; the key inequality reduces exactly to the statement that an explicit cubic polynomial has no zero in the closed unit disc, which is certified by a kernel-checked quadtree. (ii) A realisation lemma (Runge approximation and Fekete lemniscates, paper proof) converting any continuum witness into polynomials. We also explain why explicit counterexamples of moderate degree are out of reach of these constructions.

1 Introduction

Smale's mean value problem [1] asks for the best constant K such that for every complex polynomial p of degree $d \geq 2$ and every $z \in \mathbb{C}$ with $p'(z) \neq 0$ there is a critical point c of p with

$$\frac{|p(z)-p(c)|}{|z-c|} \;\leq\; K\,|p'(z)|. \tag{1.1}$$

Smale proved (1.1) with $K = 4$ and conjectured $K = 1$, or even $K = (d-1)/d$, which is attained by $z^d - dz$ at $z = 0$. The constant 4 has been improved by degree-dependent amounts that tend to 0 as $d \to \infty$: Beardon–Minda–Ng [2] ($4^{1-1/d}$), Fujikawa–Sugawa [3] and Conte–Fujikawa–Lakic [4] (improvements of order $1/d$, e.g. $4(d-1)/(d+1)$ in [4]), and Crane [5], whose improvement is of order $1/\sqrt{d}$ ($4 - 2.263/\sqrt{d}$ for $d \geq 8$); Dubinin [7] solved Smale's problem for polynomials $z^n + \cdots + c_1 z$ as a function of $|c_1|$. The conjecture was verified in low degree: Tischler [8] for $d \leq 4$, and $d = 5$ by Crane with interval arithmetic (announced in [6], "A computational proof of the degree 5 case of Smale's mean value conjecture"); degree 6 is described as the first open case in [11]. Extensive numerical searches, including AlphaEvolve as reported in [10], found no violation of $(d-1)/d$.

In September 2026 a Lean-verified disproof of the $K = 1$ form was released [9]: there is a polynomial with $p(0) = 0$, $p'(0) = 1$ and $|p(c)/c| > 1$ at all critical points c. Its degree is "very large, unspecified" and the violation is "by a small margin". With the parameters of [9] (lemniscate $|z^4 + 64i| \le \sqrt{4097}$, base point $33/32 + 2i$, Blaschke parameter $1 - t^2$) the margin of the underlying continuum construction is about $1 + 3 \cdot 10^{-7}$ (our computation). As far as we know (October 14, 2026) no quantitative lower bound $K_d \ge m > 1$ has been published.

What is new. The qualitative part of Theorem 1.1 below (failure of Smale's conjecture in *all* sufficiently large degrees) already follows from [9] together with our monotonicity Lemma 3.2. Moreover, Steps 4–5 of our realisation Lemma 3.1 (a base polynomial plus the primitive of a high power of a lemniscate polynomial, whose critical points are forced into a thin collar) follow the prototype `eventually_allBadWitness_perturbation` of the GPT-6 Astra proof [9], which uses the primitive of q^{2n} on the lemniscate $\{|q| \le 1\}$, $q = (z^4 + 64i)/\sqrt{4097}$. The new contributions are: the fjord mechanism giving continuum witnesses whose margin is bounded away from 1; an explicit witness with margin > 1.048 whose defining properties are formally verified; the general realisation lemma for arbitrary witnesses; and hence the explicit quantitative bound $K_\infty \ge 1.048$.

Result. Let $K_\infty = \sup_d K_d$.

Theorem 1.1. *$K_d \le K_{d+1}$ for all $d \ge 2$, and $K_\infty = \lim_d K_d \ge 1.048$ (indeed ≥ 1.048069073225, see Remark 2.3). Consequently there is d_0 such that for all $d \ge d_0$ some polynomial of degree d violates (1.1) with $K = 1.048$; in particular both the $K = 1$ and the $(d-1)/d$ forms of Smale's conjecture fail in all degrees $d \ge d_0$.*

The number d_0 is effective in principle but astronomically large; we do not compute it.

2 Continuum witnesses

Definition 2.1. A *continuum witness of margin m* is a triple (D, z_0, F) where D is a Jordan domain with analytic boundary, $z_0 \in D$, F is holomorphic on a neighbourhood of $\overline{D}$, $F(z_0) = 0$, $F'(z_0) = 1$, $F' \ne 0$ on $\overline{D}$, and $|F(z)| \ge m|z - z_0|$ for $z \in \partial D$.

The Blaschke mechanism (implicit in [9]). Let Φ be analytic and univalent on a neighbourhood of $\overline{\mathbb{D}}$, $D = \Phi(\mathbb{D})$, $w_0 \in \mathbb{D}$, $z_0 = \Phi(w_0)$, $a \in \mathbb{D}$, $B(w) = (w - a)/(1 - \bar{a}w)$, and define F on $\overline{D}$ by

$$F(\Phi(w)) = H(w) := \big(\Phi(w) - z_0\big)\frac{B(w)}{B(w_0)}.$$

Since $|B| = 1$ on $\partial\mathbb{D}$, $|F(z)| = |z - z_0|/|B(w_0)|$ exactly on ∂D: the margin is $m = 1/|B(w_0)|$. Moreover $F(z_0) = 0$, $F'(z_0) = H'(w_0)/\Phi'(w_0) = 1$, and $F' \ne 0$ on $\overline{D}$ iff

$$H'(w) = \frac{\Phi'(w)B(w) + (\Phi(w) - z_0)B'(w)}{B(w_0)} \ne 0 \qquad (|w| \le 1). \tag{2.1}$$

For D star-shaped with respect to z_0 a single Blaschke factor never satisfies (2.1) (argument principle). The construction needs a narrow fjord of the complement, with z_0 and the zero $\Phi(a)$ of F on opposite sides of the end of the fjord; the zero then creates a pair of critical points of F which, for suitable parameters, lie just outside $\overline{D}$.

The rounded Koebe witness. Let $k(u) = u/(1-u)^2$ (the Koebe function), $0 < r < 1$, $|c| < 1$, $u(w) = r(w+c)/(1+\bar{c}w)$ and

$$\Phi(w) = k(u(w)) = \frac{r(w+c)(1+\bar{c}w)}{E(w)^2}, \qquad E(w) = (1+\bar{c}w) - r(w+c).$$

u maps $\overline{\mathbb{D}}$ into the disc $|u| \le r$, so Φ is a rational function, analytic and univalent on a neighbourhood of $\overline{\mathbb{D}}$ ($k(u_1) = k(u_2)$ forces $(u_1 - u_2)(1 - u_1u_2) = 0$), and $D = \Phi(\mathbb{D})$ is the image of a disc under the Koebe function: a large domain with a thin fjord along the negative real axis ending near $-1/4$. A direct computation gives

$$\Phi'(w) = \frac{r(1-|c|^2)S(w)}{E(w)^3}, \quad S(w) = (1+\bar{c}w)+r(w+c), \qquad H'(w)\,E(w)^3(1-\bar{a}w)^2B(w_0) = P(w),$$

$$P(w) = r(1-|c|^2)S(w)(w-a)(1-\bar{a}w) + (1-|a|^2)\big(r(w+c)(1+\bar{c}w)E(w) - z_0E(w)^3\big), \quad (2.2)$$

a *cubic* polynomial. Hence condition (2.1) is equivalent to: *P has no zero in the closed unit disc.* We take the Gaussian-rational parameters

$$r = \tfrac{497498487}{500000000}, \qquad c = -\tfrac{105117631}{125000000} + \tfrac{3016503}{10^9}\,i,$$
$$w_0 = -\tfrac{149268011}{200000000} + \tfrac{313274719}{10^9}\,i, \qquad a = -\tfrac{744989837}{10^9} - \tfrac{504190327}{10^9}\,i.$$

Then $z_0 = \Phi(w_0) \approx -0.2500528 + 0.0001127\,i$ lies just above the end of the fjord and the second zero $\Phi(a) \approx -0.2501394 - 0.0000931\,i$ of F just below it (Figure 1); the zeros of P have moduli 1.1612, 1.0001069, 1.0000986; and $m = 1/|B(w_0)| = 1.048069073225203141728\ldots$.

Theorem 2.2 (formally verified in Lean 4 / Mathlib)**.** *For these parameters Φ is injective with $\Phi' \neq 0$ on $\overline{\mathbb{D}}$, $H' \neq 0$ on $\overline{\mathbb{D}}$, $|H(w)| = |\Phi(w) - \Phi(w_0)|/|B(w_0)|$ for $|w| = 1$, $|w_0| < 1$, $H(w_0) = 0$, $H'(w_0) = \Phi'(w_0)$, and $1/|B(w_0)| > 1.048$. Consequently $(\Phi(\mathbb{D}), \Phi(w_0), F)$ is a continuum witness of margin $m > 1.048$.*

Proof sketch of the formal proof. Injectivity: the identity $k(u_1)-k(u_2) = (u_1-u_2)(1-u_1u_2)/((1-u_1)^2(1-u_2)^2)$, the bound $|w+c| \le |1+\bar{c}w|$ for $|w| \le 1$ (from $|1+\bar{c}w|^2-|w+c|^2 = (1-|w|^2)(1-|c|^2)$) and $r < 1$. Derivatives: product/quotient rules and the polynomial identity (2.2) (`field_simp`, `ring`). Zero-freeness of P: P is computed exactly in Gaussian-rational arithmetic and a quadtree of 561 nodes covering $[-1,1]^2$ is checked by kernel evaluation; every leaf box either misses the closed disc or satisfies $|T_0| > \sum_{k\ge1}(|\Re T_k|+|\Im T_k|)\,d^k$, where $P(\zeta+\delta) = \sum_k T_k\delta^k$ is the exact Taylor expansion at the box centre ζ and $d = \frac{3}{2}h \ge \sqrt{2}\,h$ bounds $|\delta|$ on a box of half-width h. Boundary identity: $|B| = 1$ on the unit circle. Margin: the rational inequality $|w_0 - a|^2\,(131/125)^2 < |1 - \bar{a}w_0|^2$. □

From the Lean statement to Definition 2.1. The Lean theorem is stated in the coordinate w. Put $D = \Phi(\mathbb{D})$, $z_0 = \Phi(w_0)$ and $F = H \circ \Phi^{-1}$ on $\Phi(\{|w| < \rho_*\})$, where $\rho_* = 1.00043$: for $|w| \le \rho_*$ the identity $|1+\bar{c}w|^2 - |w+c|^2 = (1-|w|^2)(1-|c|^2)$ and $|1+\bar{c}w| \ge 1 - |c|\rho_*$ give $r^2|w+c|^2 \le r^2|1+\bar{c}w|^2 + r^2(\rho_*^2-1)(1-|c|^2) < |1+\bar{c}w|^2$ (an exact rational check with $|c| \le 0.840947$), so u maps $\{|w| < \rho_*\}$ injectively into $\mathbb{D}$, where k is univalent; hence Φ is analytic and univalent on $|w| < \rho_*$, D is a Jordan domain with analytic boundary $\Phi(\partial\mathbb{D})$ and F is holomorphic on the neighbourhood $\Phi(\{|w| < \rho_*\})$ of $\overline{D}$. By the chain rule $F'(\Phi(w)) = H'(w)/\Phi'(w)$, so the Lean statement gives $F(z_0) = H(w_0) = 0$, $F'(z_0) = H'(w_0)/\Phi'(w_0) = 1$, $F' \neq 0$ on $\overline{D}$, and for $z = \Phi(w) \in \partial D$: $|F(z)| = |H(w)| = |z - z_0|/|B(w_0)| = m|z - z_0|$ with $m > 1.048$.

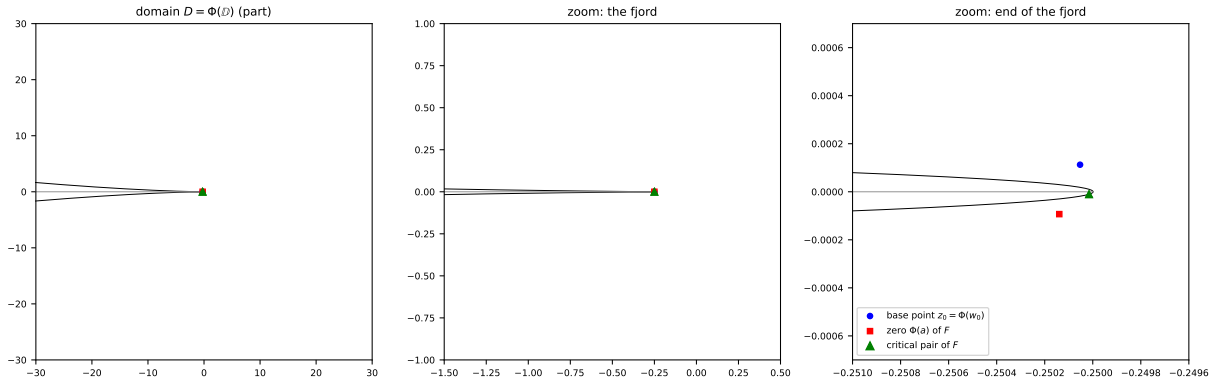


Figure 1: The Koebe witness: the domain $D = \Phi(\mathbb{D})$ (only a part is shown; D extends to $\Re z \approx 4 \cdot 10^4$), its fjord along the negative axis, and the end of the fjord with the base point z_0, the second zero $\Phi(a)$ of F and the critical pair of F (green), which lies just outside $\overline{D}$.

Remark 2.3 (exact margin). Lean certifies $m = 1/|B(w_0)| > 131/125 = 1.048$. The sharper statement $1.048069073225 < m < 1.048069073226$ is an exact rational comparison of $m^2 = |1 - \bar{a}w_0|^2/|w_0 - a|^2$ with the squares of these decimals, done outside Lean (Python `fractions`; also in the referee script `data/check_witness.py`).

Independent witnesses (ball arithmetic). Two further explicit families, rounded slit discs $D = \Phi_0(r\mathbb{D})$ with $\Phi_0 : \mathbb{D} \to \mathbb{D} \setminus [0, 1)$ and $\Phi_0 : \mathbb{D} \to \mathbb{D} \setminus ([s, 1) \cup (-1, -s])$ given by square roots and Joukowski maps, yield witnesses of margins $[1.037310491225903\ldots \pm 7 \cdot 10^{-34}]$ and $[1.041615718017113\ldots \pm 7 \cdot 10^{-34}]$; there condition (2.1) involves algebraic functions and was certified with Arb ball arithmetic [13] (adaptive arcs of the unit circle, certified winding number 0). These are not needed for Theorem 1.1 but confirm the mechanism in three different geometries; data and code are in the repository. An independent referee script (`data/check_witness.py`; exact rationals and mpmath) recomputes z_0, the cubic P, its zeros, the margin and replays the quadtree certificate.

3 From continuum witnesses to polynomials

Lemma 3.1 (Realisation)**.** *If (D, z_0, F) is a continuum witness of margin m, then for every $\varepsilon > 0$ there is a polynomial p with $p'(z_0) \neq 0$ and $R(p, z_0) \geq m - \varepsilon$.*

Proof. The constants are chosen in the order $\varepsilon \to \rho, \rho_1 \to n, \theta_0, \kappa, \Lambda \to \delta \to G \to M$; each depends only on the previous ones (and on D, z_0, F).

Step 1 (enlargement). Let Φ be analytic and univalent on $|w| < \rho_*$, $\rho_* > 1$, with $\Phi(\mathbb{D}) = D$ (for the witness of Theorem 2.2 take $\rho_* = 1.00043$, see the paragraph after its proof). Given ε, choose $1 < \rho_1 < \rho < \rho_*$ such that, with $D' = \Phi(\rho\mathbb{D})$, F is holomorphic near $\overline{D'}$, $F' \neq 0$ on $\overline{D'}$, and $|F(z)| \geq (m - \varepsilon/4)|z - z_0|$ on $\overline{D'} \setminus D$ (possible by continuity, since the strict inequality $|F(z)| > (m - \varepsilon/4)|z - z_0|$ holds on the compact set ∂D and $z_0 \notin \partial D$). Let $K = \Phi(\rho_1\overline{\mathbb{D}})$, g_K its Green function with pole at ∞, and $\gamma = \min_{\partial D'} g_K > 0$; by the minimum principle $g_K \geq \gamma$ on $\mathbb{C} \setminus D'$. Put $d_0 = \operatorname{dist}(z_0, \partial D) > 0$.

Step 2 (lemniscate; choice of $n, \theta_0, \kappa, \Lambda$). Let q_n be monic Fekete polynomials of K (zeros at Fekete points, on ∂K). Then $\|q_n\|_K^{1/n} \to \operatorname{cap} K$ and $\frac{1}{n}\log|q_n| \to g_K + \log\operatorname{cap} K$ locally uniformly on $\mathbb{C}\setminus K$ [12, §5.5]. Put $Q = q_n/(\operatorname{cap} K\, e^{\gamma/2})^n$. Then $\|Q\|_K \to 0$; on the compact set $\{g_K \geq \gamma,\ |z| \leq R_0\}$ we get $|Q| \geq e^{n(\gamma/2 - o(1))}$; and for $|z| \geq R_0 := \operatorname{diam} K + 2\operatorname{cap} K\, e^{\gamma/2}$ (taking $0 \in K$ w.l.o.g.) $|Q(z)| \geq \big((|z| - \operatorname{diam} K)/(\operatorname{cap} K\, e^{\gamma/2})\big)^n \geq 2^n \geq 2$. Fix n so large that $|Q| < 1$ on K and $|Q| \geq \lambda_0 > 1$ on $\mathbb{C}\setminus D'$. All zeros of Q lie in the component of $\{|Q| < 1\}$ containing K, which by Riemann–Hurwitz then contains all $n-1$ critical points of Q. Choose $\theta_0 > 0$ with $|Q| \leq 1 - 2\theta_0$ on K, all critical values of modulus $< 1 - 2\theta_0$, and $\lambda_0 > 1 + 2\theta_0$. Then $L = \{|Q| \leq 1 - \theta_0\}$ contains all critical points of Q, hence is connected, and since $1 - \theta_0$ is not a critical value, ∂L is an analytic Jordan curve and L is a closed Jordan domain with analytic boundary; in particular any two points of L are joined by a rectifiable path in L and the intrinsic diameter Λ of L is finite. Q' has no zero on the band $S = \{1 - \theta_0 \leq |Q| \leq 1 + \theta_0\} \subset D' \setminus K$; put $\kappa = \min_S |Q'| > 0$.

Step 3 (Runge; choice of δ and G). Choose $\delta > 0$ (depending on ε, d_0, $\mu_F := \min_{\overline{D'}} |F'|$) so small that $2\delta/d_0 < \varepsilon/8$, $\delta < \mu_F/2$ and $m\delta < \varepsilon/16$, and a polynomial G with $|G - F| < \delta$ and $|G' - F'| < \delta$ on $\overline{D'}$ (Runge's theorem on a neighbourhood of $\overline{D'}$ plus Cauchy estimates). Then $\mu := \min_{\overline{D'}} |G'| \geq \mu_F/2$ and $\Gamma := \max_{\overline{D'}} |G'| < \infty$.

Step 4 (choice of M; location of the critical points). Let $p_M(z) = G(z) + \int_{z_0}^{z} Q(t)^M\, dt$. Its critical points solve $G' = -Q^M$. On $\{|Q| \leq 1 - \theta_0\} \subset D'$ we have $|Q|^M \leq (1-\theta_0)^M < \mu \leq |G'|$ for M large. On $\{|Q| \geq 1 + \theta_0\}$: on the compact part $\{|Q| \geq 1 + \theta_0,\ |z| \leq R_1\}$ we have $|Q|^M \geq (1+\theta_0)^M > \max_{|z|\leq R_1} |G'|$ for M large; and for $|z| \geq R_1 := \max(R_0, R_2)$, where R_2 is such that $|Q(z)|^{M_1} > |G'(z)|$ for $|z| \geq R_2$ with $M_1 = \deg G$ (possible because $|Q(z)| \geq c_Q|z|^n$ near ∞), we have $|Q| \geq 2$, hence $|Q|^M \geq |Q|^{M_1} > |G'|$ for all $M \geq M_1$. So for M large every critical point c lies in S; in particular $c \in D' \setminus \overline{D}$ and $|c - z_0| \geq d_0$.

Step 5 (critical values). Q restricted to $\{1 - 2\theta_0 < |Q| < 1 + 2\theta_0\}$ has no critical points and is proper onto the annulus $\{1 - 2\theta_0 < |\zeta| < 1 + 2\theta_0\}$, hence a covering. Lift the radial segment from $Q(c)$ to $Q(c)(1-\theta_0)/|Q(c)|$ through c; along the lift $1 - \theta_0 \leq |Q| \leq |Q(c)| \leq 1 + \theta_0$, so *the lifted path stays in S*, and it ends at a point $c' \in \partial L$. Parametrising by $s = |Q(z)| \in [1 - \theta_0, |Q(c)|]$ we have $dz = (Q(c)/|Q(c)|)\, ds/Q'(z)$ with $|Q'| \geq \kappa$, and $|Q(c)|^M = |G'(c)| \leq \Gamma$, so

$$\Big|\int_{c'}^{c} Q^M\Big| \leq \frac{|Q(c)|^{M+1}}{\kappa(M+1)} = \frac{|G'(c)|\,|Q(c)|}{\kappa(M+1)} \leq \frac{\Gamma(1+\theta_0)}{\kappa(M+1)}.$$

Joining z_0 to c' inside L by a path of length $\leq \Lambda$ gives $|\int_{z_0}^{c'} Q^M| \leq \Lambda(1-\theta_0)^M$. Thus $p_M(c) - p_M(z_0) = G(c) - G(z_0) + e_M$ with $|e_M| \leq \Gamma(1+\theta_0)/(\kappa(M+1)) + \Lambda(1-\theta_0)^M \to 0$, uniformly in the critical points c.

Step 6. $|G(c) - G(z_0)| \geq |F(c)| - 2\delta \geq (m - \varepsilon/4)|c - z_0| - 2\delta$ and $|p_M'(z_0)| = |G'(z_0) + Q(z_0)^M| \leq 1 + \delta + (1 - 2\theta_0)^M$. With $|c - z_0| \geq d_0$ and the choice of δ this gives $R(p_M, z_0) \geq \big(m - \varepsilon/4 - \varepsilon/8 - |e_M|/d_0\big)/(1 + \delta + (1-2\theta_0)^M) \geq m - 3\varepsilon/8 - m\delta - o(1) \geq m - \varepsilon$ for M large. □

Lemma 3.2 (Monotonicity)**.** $K_{d+1} \geq K_d$.

Proof. Let $\deg p = d$, $p(0) = 0$, $p'(0) = 1$, $R(p, 0) = \rho$. For $\varepsilon \to 0$ the polynomial $p_\varepsilon = p + \varepsilon z^{d+1}/(d+1)$ has $d-1$ critical points converging (with multiplicity) to those of p, so the corresponding ratios tend to values $\geq \rho$, and one critical point $c_\varepsilon \sim -\operatorname{lc}(p')/\varepsilon \to \infty$ with $p_\varepsilon(c_\varepsilon) \sim \operatorname{lc}(p')\, c_\varepsilon^d\big(\frac{1}{d} - \frac{1}{d+1}\big)$, whose ratio tends to ∞. Also $p_\varepsilon(0) = 0$, $p_\varepsilon'(0) = 1$. Hence $K_{d+1} \geq \rho - o(1)$. □

Proof of Theorem 1.1. By Theorem 2.2 and Lemma 3.1, for every $\varepsilon > 0$ some $K_d \geq 1.048069 - \varepsilon$; by Lemma 3.2 K_d is non-decreasing, so $\lim K_d = \sup K_d \geq 1.048069$. Taking d_0 with $K_{d_0} > 1.048$ gives the last statement. □

4 Why explicit counterexamples are hard

Remark 4.1 (first-order obstruction; heuristic). If N critical points are placed on a smooth curve ∂D around 0 according to its equilibrium measure (so that $p' \approx 1$ inside), a saddle-point computation gives $R \approx 1 - \max_{c\in\partial D} \mathrm{Re}\big(1/\kappa(c)\big)/N$ with $\kappa = z\Psi'/\Psi$, Ψ the exterior conformal map. Since $1/\kappa$ is holomorphic on $\widehat{\mathbb{C}} \setminus \overline{D}$ with value 1 at ∞, the maximum principle gives $\max \mathrm{Re}(1/\kappa) \geq 1$, with equality only for discs centred at 0: to first order nothing beats $(d-1)/d$. A counterexample needs an $O(1)$ design margin as in Theorem 2.2.

Remark 4.2 (discretisation obstruction; heuristic). To make $p' \approx F'$ in D with all critical points near ∂D, their density must be $N\mu_\infty + \sigma \geq 0$, where σ is an N-independent signed measure. At the end of a narrow fjord μ_∞ is exponentially small while $\sigma < 0$ nearby, so no moderate N works; equivalently, polynomial approximation of F on $\overline{D}$ converges extremely slowly (for a slit-disc witness the uniform error of degree-320 least-squares approximation on ∂D is still 1.7). Wider fjords help discretisation but cost margin: for the widest-fjord feasible design we found (margin 1.0065) a boundary-integral computation of μ_∞ and σ gives the necessary condition $N \gtrsim 4 \cdot 10^4$. Sequential linear programming directly on the critical points (up to $N = 160$, many starts) always converged to configurations with $R < (d-1)/d$.

5 Verification

Lean. Theorem 2.2 is the theorem `SmaleKoebe.witness` in `lean/Main.lean` (Lean 4.35.0, Mathlib v4.35.0; about 100 s and 8 GB). Its statement, with $\Phi(x) = r(x+c)(1+\bar{c}x)/((1+\bar{c}x) - r(x+c))^2$, $B(x) = (x-a)/(1-\bar{a}x)$, $H(x) = (\Phi(x) - \Phi(w_0))B(x)/B(w_0)$ and the explicit derivatives Φ', H', reads:

```
theorem witness :
    Set.InjOn Φ (Metric.closedBall 0 1) ∧
    (∀ w : ℂ, ‖w‖ ≤ 1 → HasDerivAt Φ (Φ' w) w ∧ Φ' w ≠ 0) ∧
    (∀ w : ℂ, ‖w‖ ≤ 1 → HasDerivAt H (H' w) w ∧ H' w ≠ 0) ∧
    (∀ w : ℂ, ‖w‖ = 1 → ‖H w‖ = ‖Φ w - Φ w0C‖ / ‖B w0C‖) ∧
    ‖w0C‖ < 1 ∧ H w0C = 0 ∧ H' w0C = Φ' w0C ∧ ((m0 : ℝ) < 1 / ‖B w0C‖)
```

with $m_0 = 131/125 = 1.048$. The command `#print axioms` reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`; there is no `sorry` and no `native_decide`. The finite computations (exact cubic, quadtree, rational side conditions) are evaluated by the kernel (`decide +kernel`).

Ball arithmetic and independent checks. The two independent witnesses are certified by `verify_cont.py` (python-flint 0.9.0 / Arb, 120 bits). The referee script `check_witness.py` (in `data/`) independently recomputes the Koebe witness in exact rational arithmetic and mpmath and replays the quadtree certificate.

Not verified formally. Lemma 3.1 (standard potential theory; paper proof) and Lemma 3.2 (paper proof); hence Theorem 1.1 is certified by Lean for its key analytic input and by paper proofs for the two general lemmas.

6 Open questions

What is K_∞? Our single-zero designs plateau near 1.048 (Koebe), 1.042 (doubly slit disc) and 1.037 (slit disc); second Blaschke zeros and a k-fold symmetric variant did not help in our experiments. What is the least degree d with $K_d > 1$, and is there an explicit example? Making the constants in Lemma 3.1 effective for a wide-fjord witness would give a compactly described explicit polynomial $p = G + \int Q^M$, of degree roughly estimated at 10^6–10^7.

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